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PHNOM PENH INTERNATIONAL UNIVERSITY

Customer Segmentation Using RFM Analysis and Unsupervised Clustering

A comparative study of K-Means, Hierarchical Clustering, and Gaussian Mixture Models on 49,661 e-commerce customers.

Lecturer: Mr. Him Soklong
Made by: Chang Tong

ID: CSE2024220035

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1. Context
2. Data Preparation
3. Data Exploring
4. Feature Engineering
5. Approach
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10. Reflection



01 • CONTEXT

Problem Statement & Dataset

The Problem

E-commerce firms have vast transaction data, but lack the behavioral insights needed for targeted retention and marketing.

The Goal

Segment ~50K customers using unsupervised RFM clustering to unlock actionable behavioral insights.

Why Feature Engineering

Data is transaction-level; behavioural signal only emerges once records are aggregated to the customer level.

DATASET

E-Commerce Customer for Behavior Analysis

Source • Kaggle

250,000

transaction records

49,661

unique customers

13

columns

2020 – 2023

Jan '20 → Sep '23

KEY COLUMNS

Customer ID • Purchase Date • Product Category • Price • Quantity • Total Amount • Payment Method • Age • Returns • Gender • Churn

02 • PREPARATION

Data Cleaning Summary

- 01 No duplicates**
Zero exact duplicate rows detected across all 250K records.
- 02 Missing 'Returns'**
≈ 19% missing values imputed as zero (no return recorded).
- 03 Invalid transactions**
Rows with zero or negative Quantity, Price, or Total Amount removed.
- 04 Redundant column**
Dropped 'Age' (exact duplicate of 'Customer Age').
- 05 Result**
Clean transaction-level table ready for customer aggregation.



03 • EXPLORATION

EDA on Raw Transactions

Key Observations

Symmetric distributions

Price, Quantity, and Total Amount are near-symmetric ($\text{skew} \approx 0$) — a strong hint of synthetic data.

Balanced categories

Product categories split $\sim 25\%$ each: Electronics, Clothing, Home, Books.

Balanced payment mix

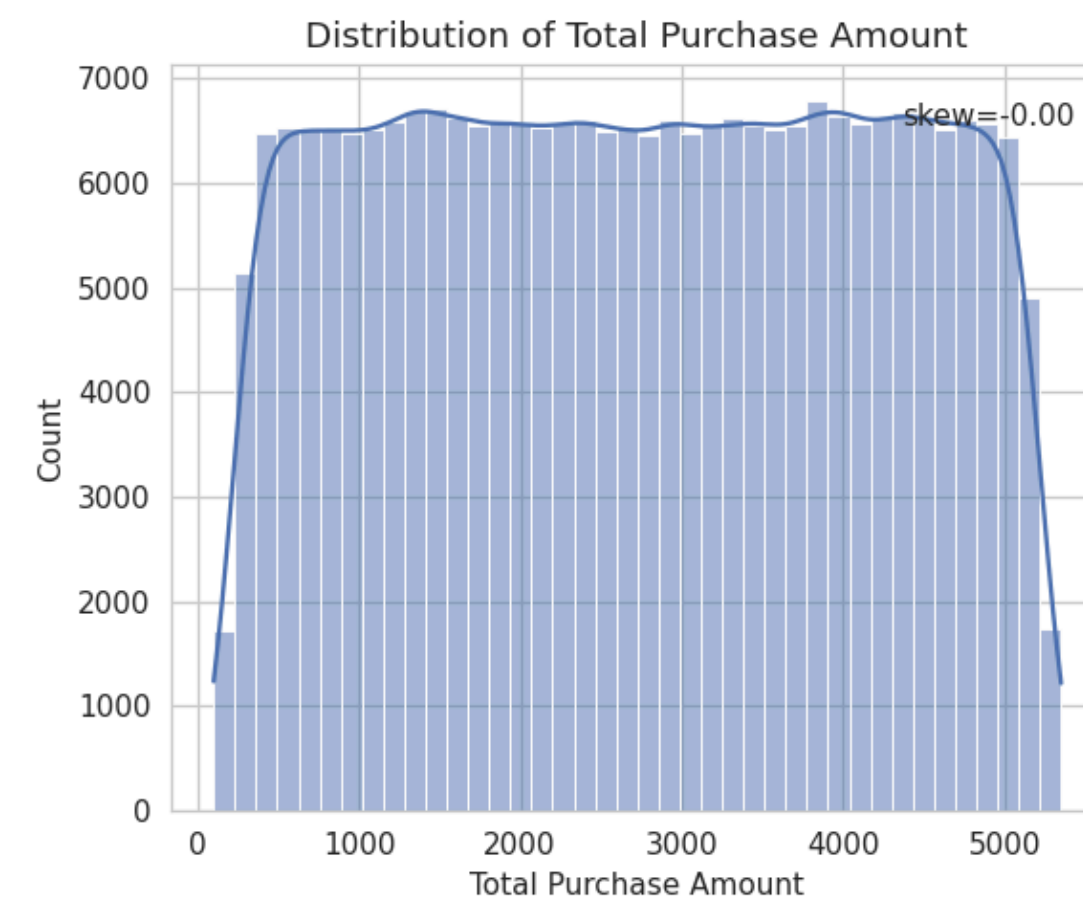
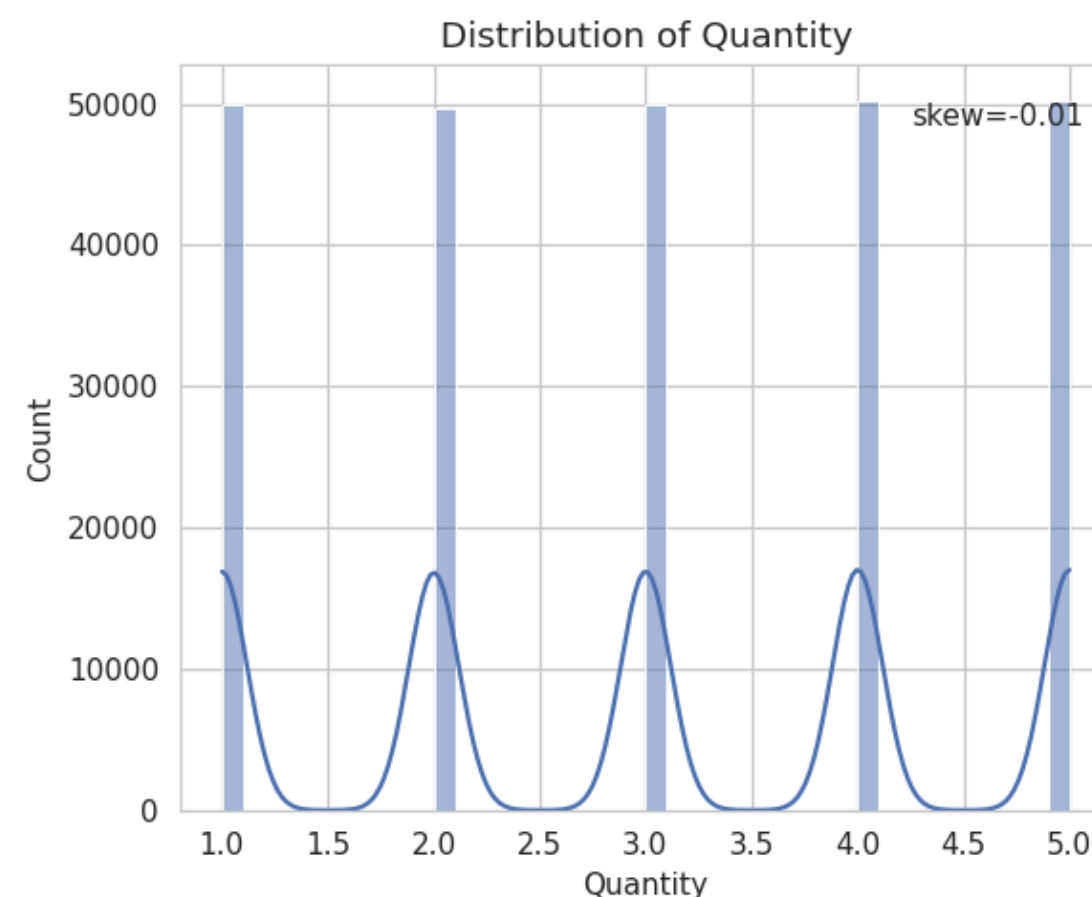
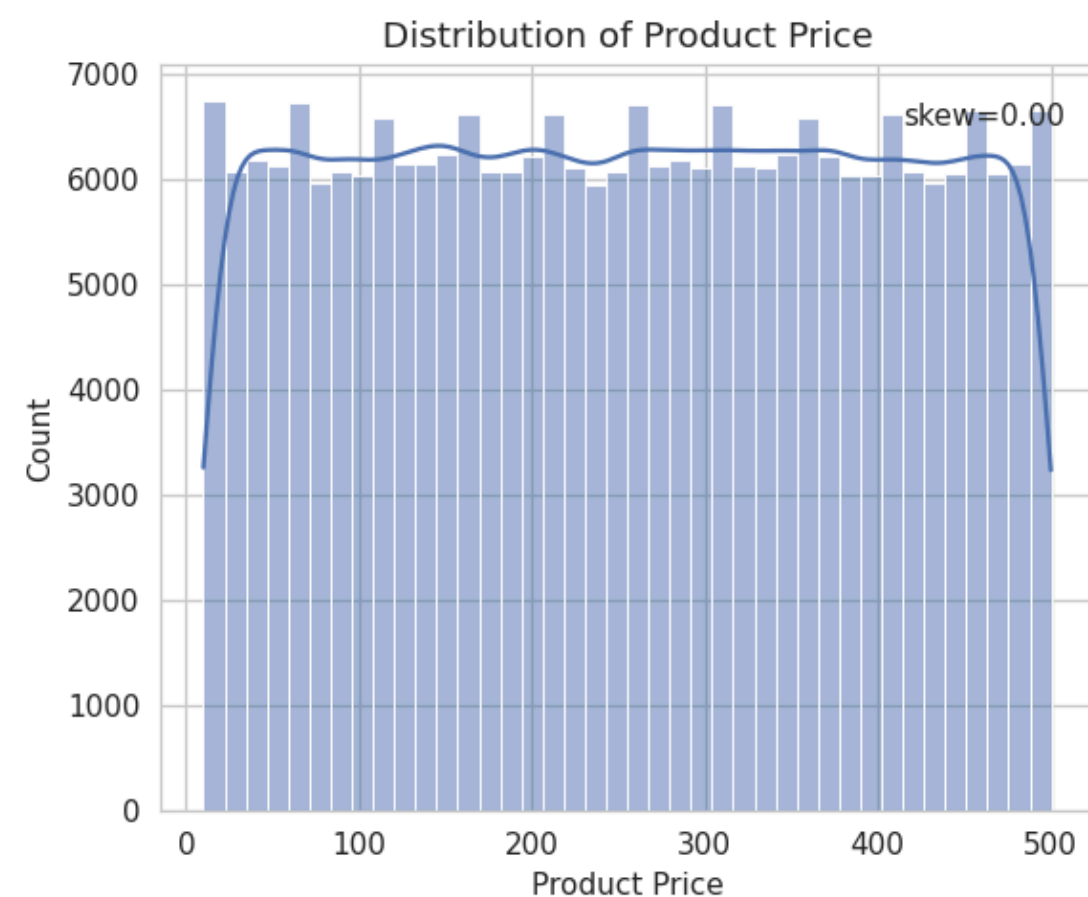
Credit Card, PayPal, and Cash roughly balanced across transactions.

Demographics

Gender split $\approx 50/50$ · average age ≈ 43.8 years.

Behavioural signals

Churn 20.1% · transaction-level return rate $\approx 50\%$.



03 • EXPLORATION

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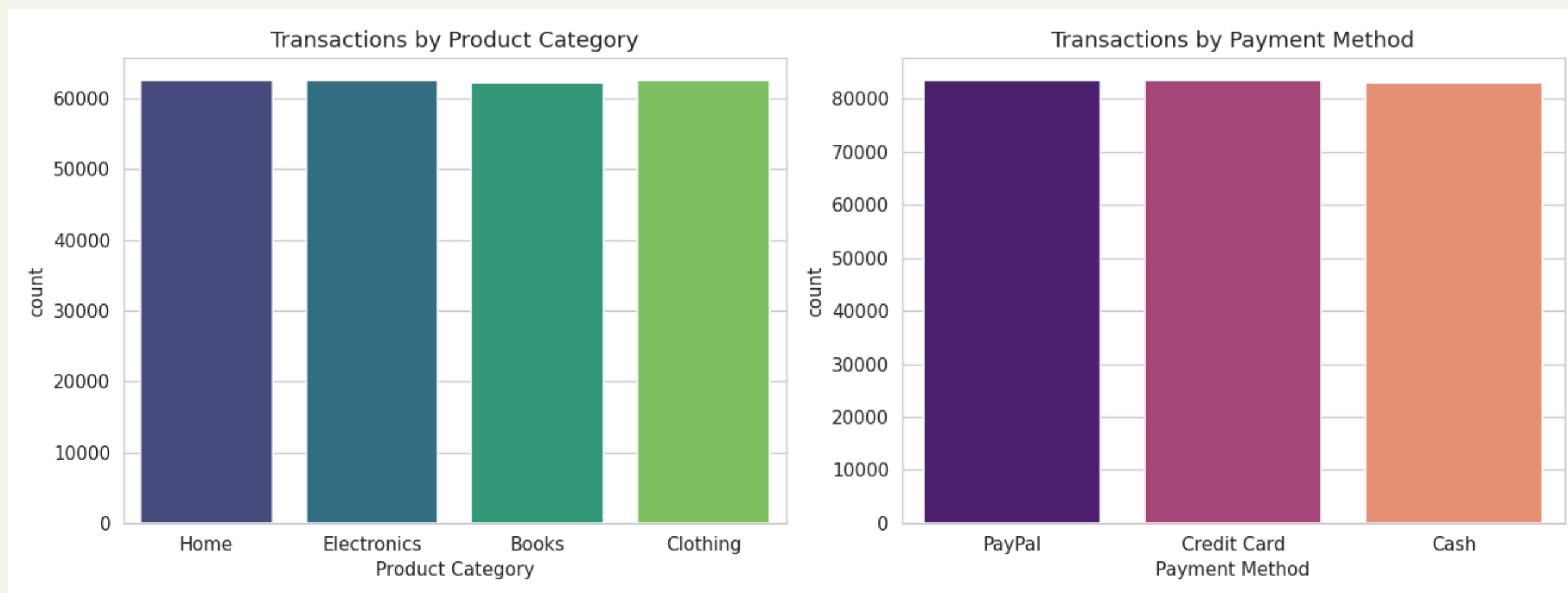
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03 • EXPLORATION

Correlation Insights

Correlation Heatmap — Transaction-Level Features



Near-zero everywhere.

01

Weak numeric correlation

All pairwise correlations between Price, Quantity, Age, Returns, and Churn hover near zero at the transaction level.

02

Raw features aren't enough

Individual transaction features do not, on their own, reveal customer behaviour patterns.

03

Justifies aggregation

Motivates aggregation to customer-level RFM features to surface meaningful behavioural signal for clustering.

04 • FEATURE ENGINEERING

RFM Feature Engineering Method

Feature that used

Recency

Frequency (log)

Monetary (log)

AOV

Return Rate %

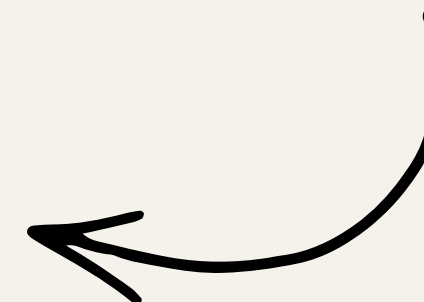
TRANSFORMATION PIPELINE

1
Aggregate → Customer

2
Log-transform skew

3
IQR outlier cap

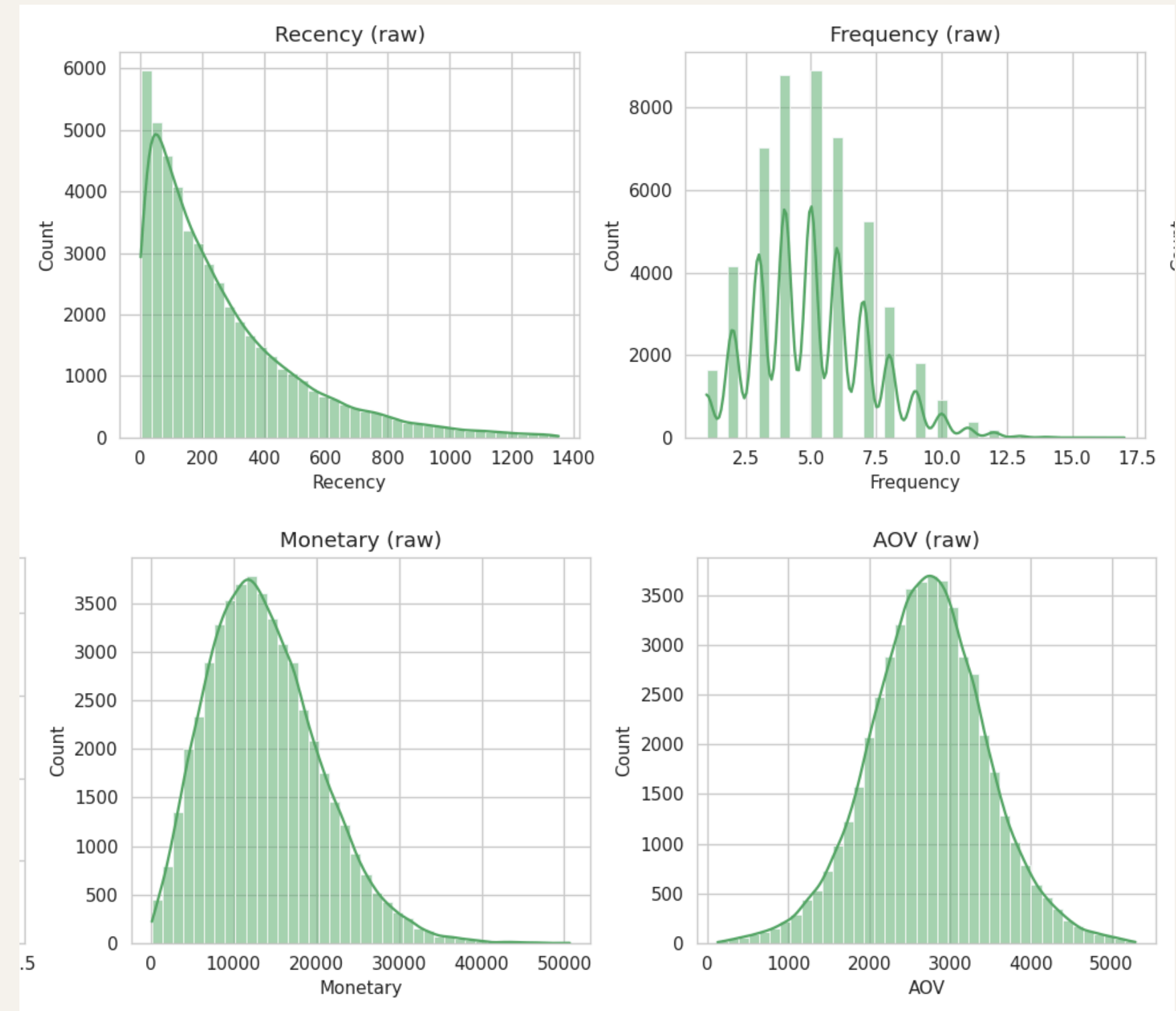
4
Standardize ($\mu=0$, $\sigma=1$)



04 • FEATURE ENGINEERING

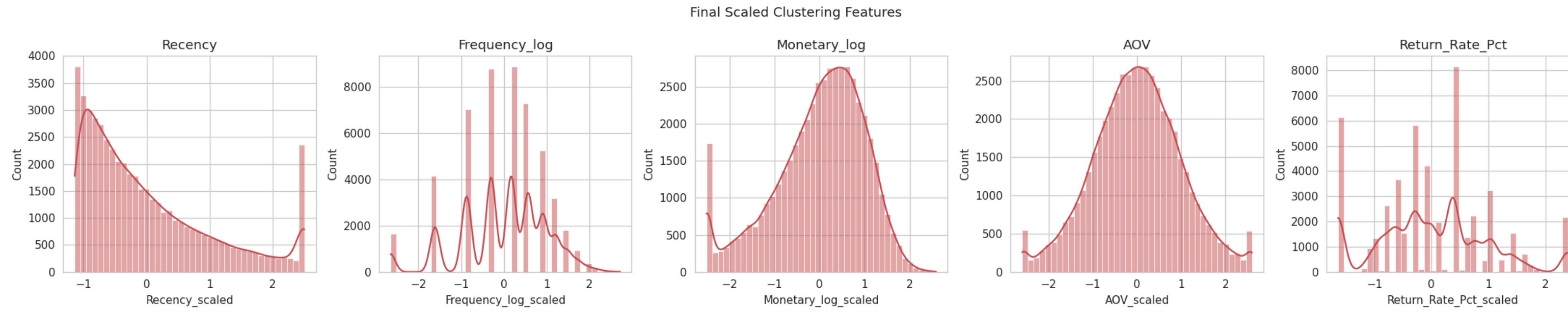
RFM Feature Engineering & Distributions

Statistic	Value
Count	49,661
Mean	\$13,720
Median	\$13,008
Std Dev	\$6,811
Min	\$125
Max	\$50,659
Skewness	0.62
Outliers (Z > 3)	298 (0.6%)



04 • FEATURE ENGINEERING

RFM Feature Engineering & Distributions



Feature	Mean	Std Dev	Min	Max	Skewness	Outliers (Z >3)
Recency_scaled	0	1	-1.14	2.5	1.05	0 (0%)
Frequency_log_scaled	0	1	-2.65	2.72	-0.51	0 (0%)
Monetary_log_scaled	0	1	-2.52	2.59	-0.64	0 (0%)
AOV_scaled	0	1	-2.60	2.6	0	0 (0%)
Return_Rate_Pct_scaled	0	1	-1.64	2.4	0.28	0 (0%)

04 • FEATURE ENGINEERING

RFM Feature Engineering finding

Customer-Level Summary

Averages across 49,661 customers after feature engineering.

254 days

Avg Recency

5 orders

Avg Frequency

\$13,720

Avg Monetary

\$2,727

Avg AOV

40.6%

Avg Return Rate



05 • APPROACH

Clustering Feature Set & Approach

01**FEATURES Selected**

Recency · Frequency (log) ·
Monetary (log) · AOV ·
Return Rate %
All standardized.

02**PARADIGM**

Unsupervised — no ground-truth labels. Algorithms discover latent segments from behaviour alone.

03**TUNING**

Grid search across k, initialization, linkage, and covariance types. Consistent metrics across methods.

04**PRIORITY**

Balanced focus on scalability to ~50K and sampling 10k customers and interpretability of resulting personas.

06 • ALGORITHMS

Algorithms Selected & Rationales

ALGORITHM 01

K-Means

- Fast, scalable to the full 49,661 customers
- Industry-standard baseline for RFM segmentation
- Tuned $k = 2-8 \cdot \text{init methods} \cdot n_{\text{init}}$

BEST CONFIGURATION

$k = 4 \cdot \text{init} = \text{random} \cdot n_{\text{init}} = 20$

ALGORITHM 02

Hierarchical

- Captures non-spherical clusters; no fixed k initially
- Useful for validating cluster count across methods
- Tuned $k = 2-8 \cdot \text{linkage} \cdot \text{distance metric}$

BEST CONFIGURATION

$k = 4 \cdot \text{linkage} = \text{complete} \cdot \text{metric} = \text{euclidean}$

CAVEAT

Ran on 4,000-customer sample due to $O(n^2)$ complexity.

ALGORITHM 03

GMM

- Soft, probabilistic assignment
- Models elliptical (non-isotropic) clusters
- Tuned $n_{\text{components}} = 2-8 \cdot \text{covariance types}$

BEST CONFIGURATION

$n_{\text{components}} = 4 \cdot \text{covariance} = \text{tied}$

06 • ALGORITHMS

Algorithms Selected & Rationales

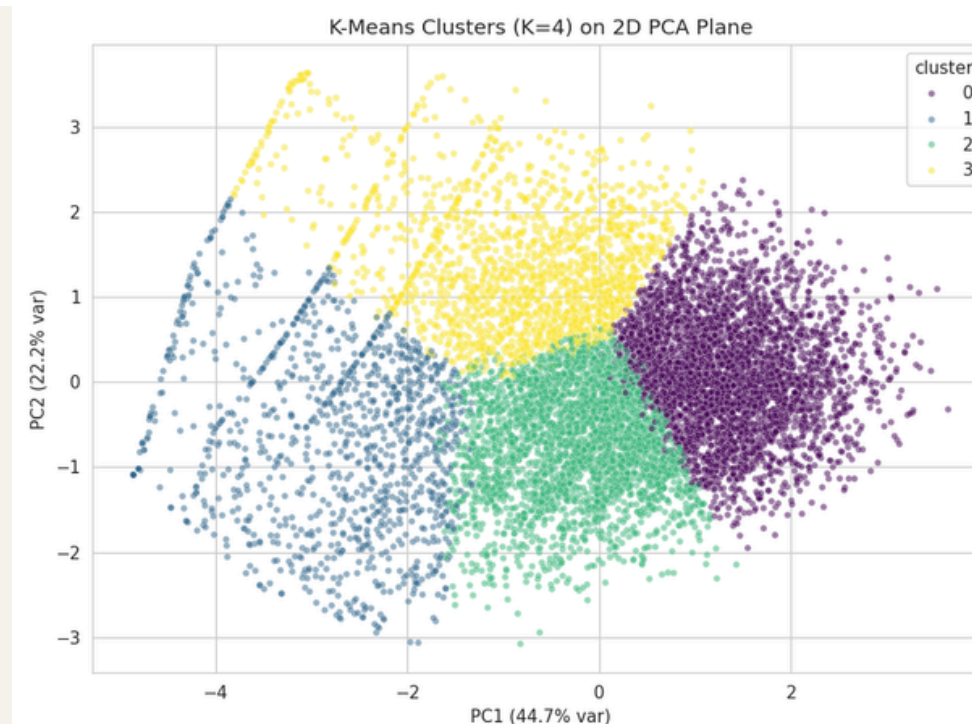
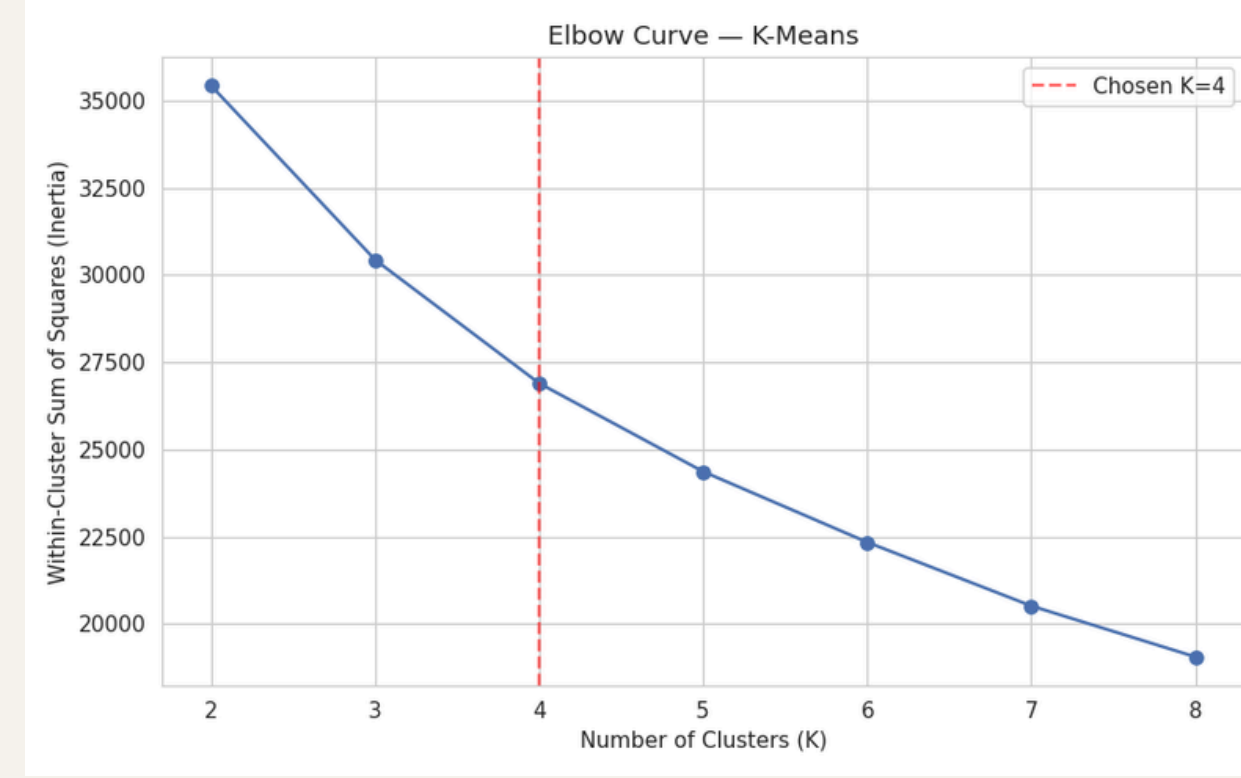
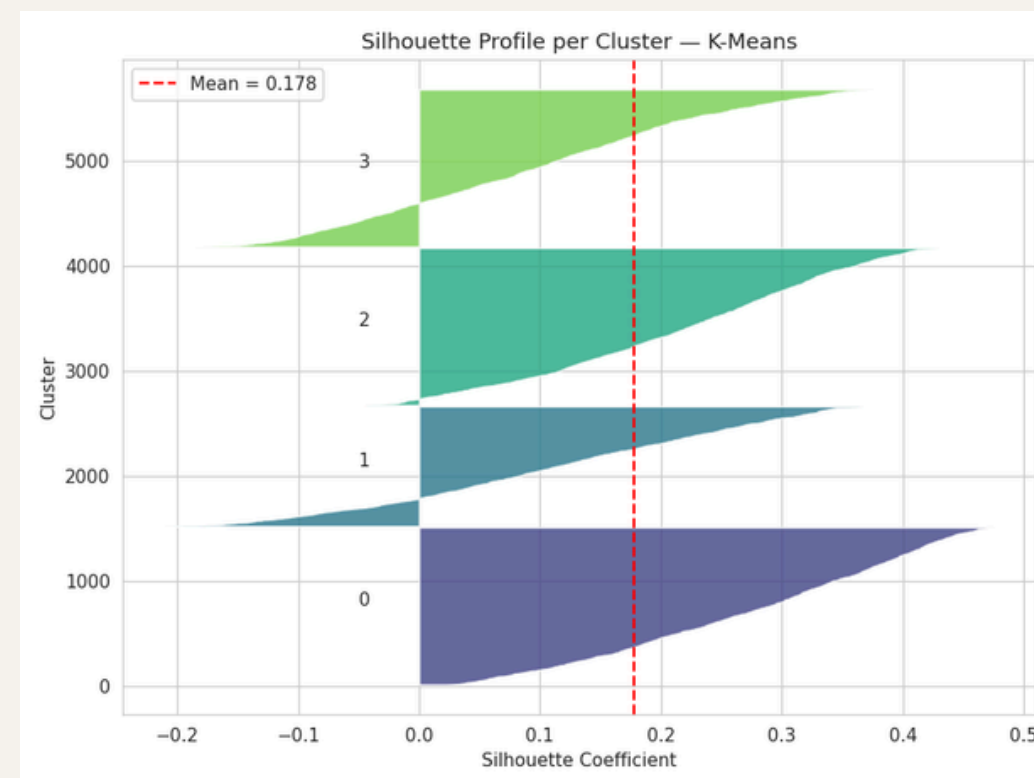
ALGORITHM 01

K-Means

- Fast, scalable of 10k out of 50k customers
- Industry-standard baseline for RFM segmentation
- Tuned $k = 2-8$ · init methods · n_{init}

BEST CONFIGURATION

$k = 4$ · init = random · $n_{init} = 20$

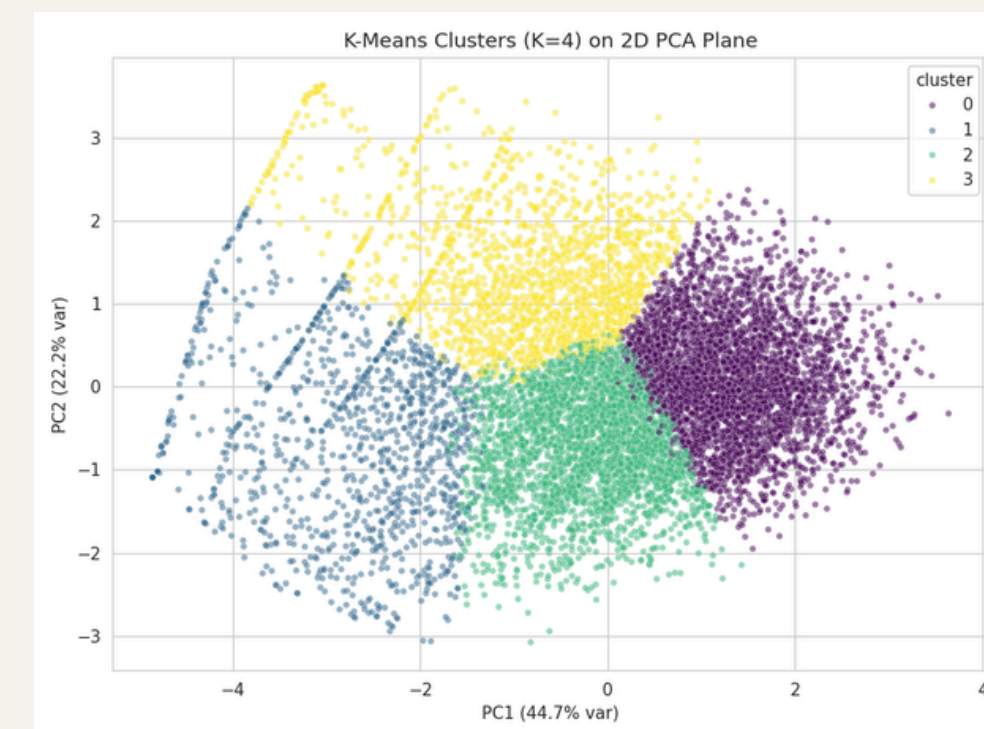
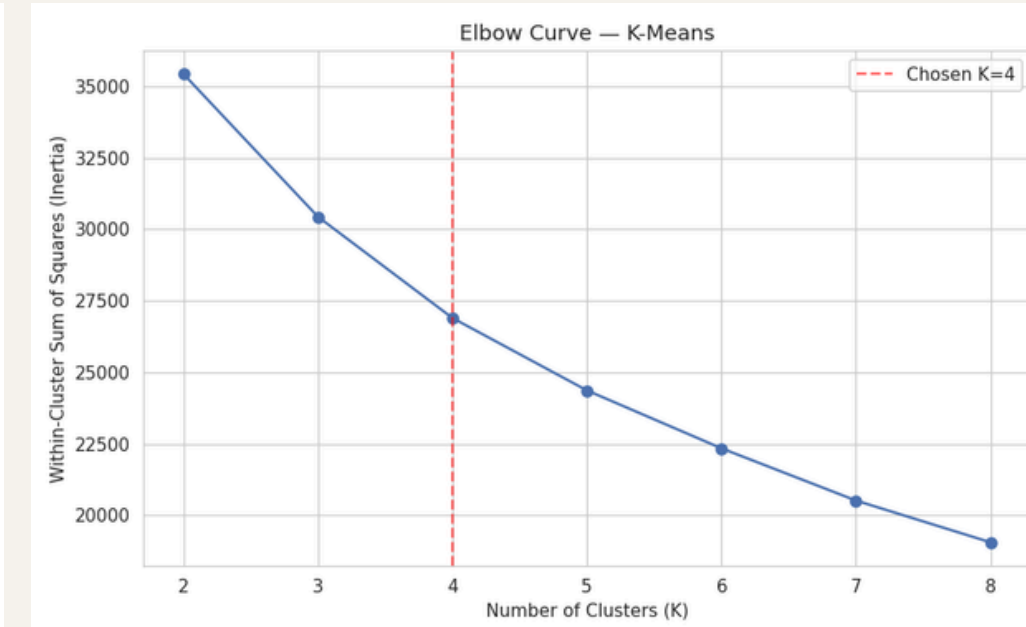
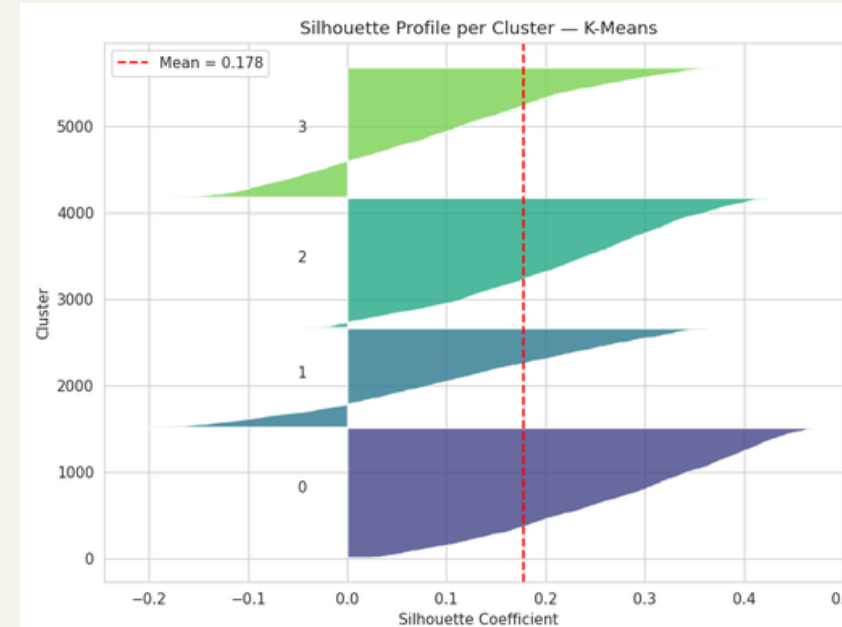


06 • ALGORITHMS

Algorithms Selected & Rationales

Result and Interpretation

- K=4 selected via Elbow curve (init=random, n_init=20).
- Metrics: Silhouette = 0.247 | Davies-Bouldin = 1.4693 | Calinski-Harabasz = 2868.2
- Trade-off: K=2 had higher Silhouette (0.3054), but K=4 yields balanced, actionable personas.
- Next Steps: K-Means' spherical assumption struggles with irregular data, motivating tests with GMM and Hierarchical Clustering.



06 • ALGORITHMS

Algorithms Selected & Rationales

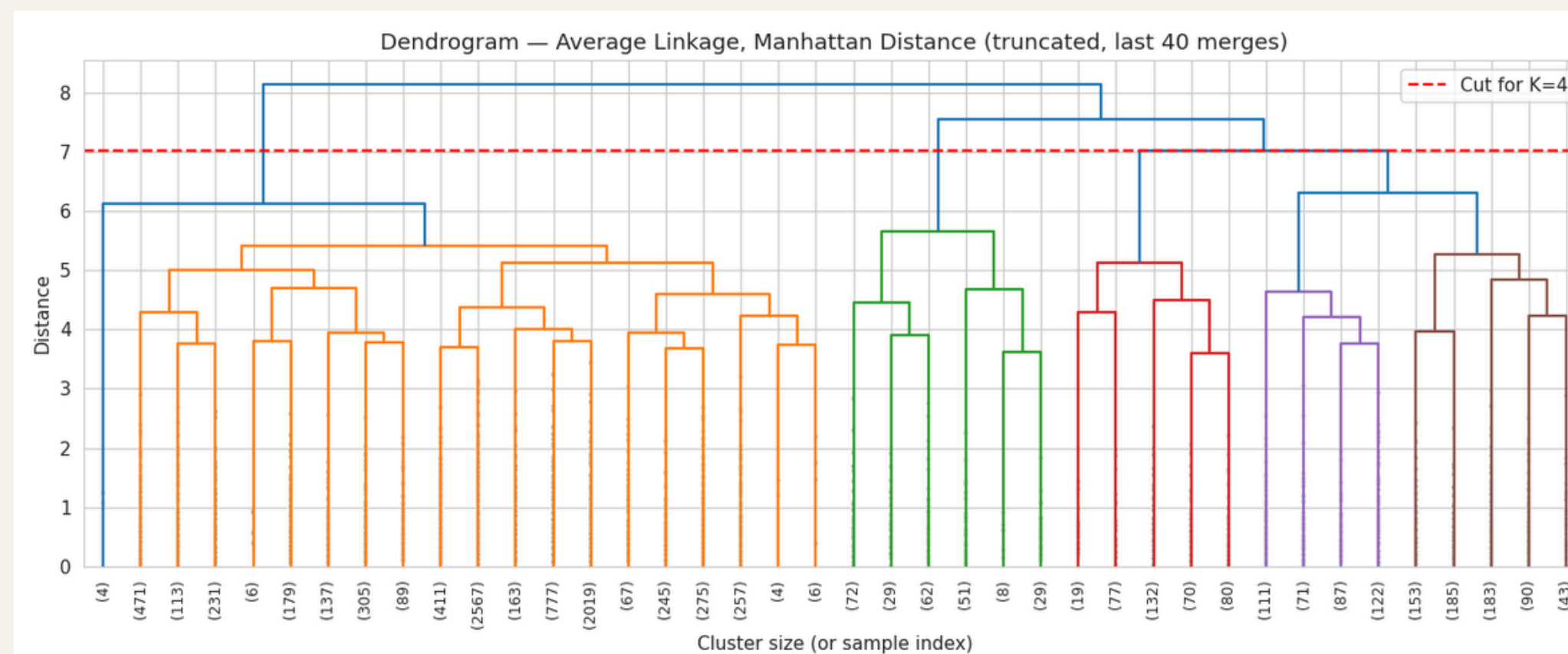
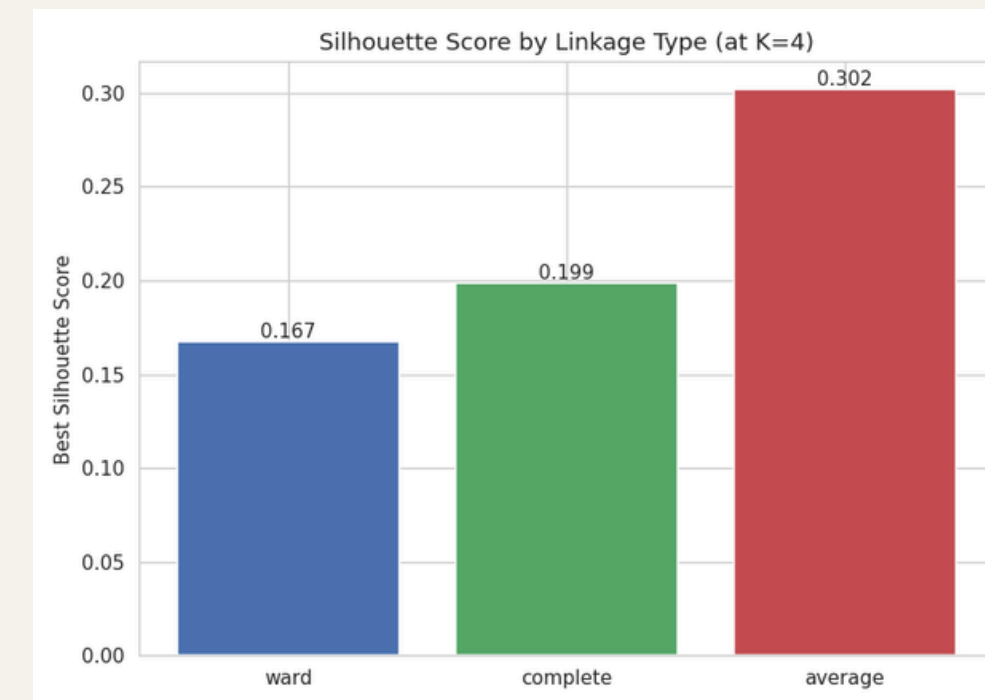
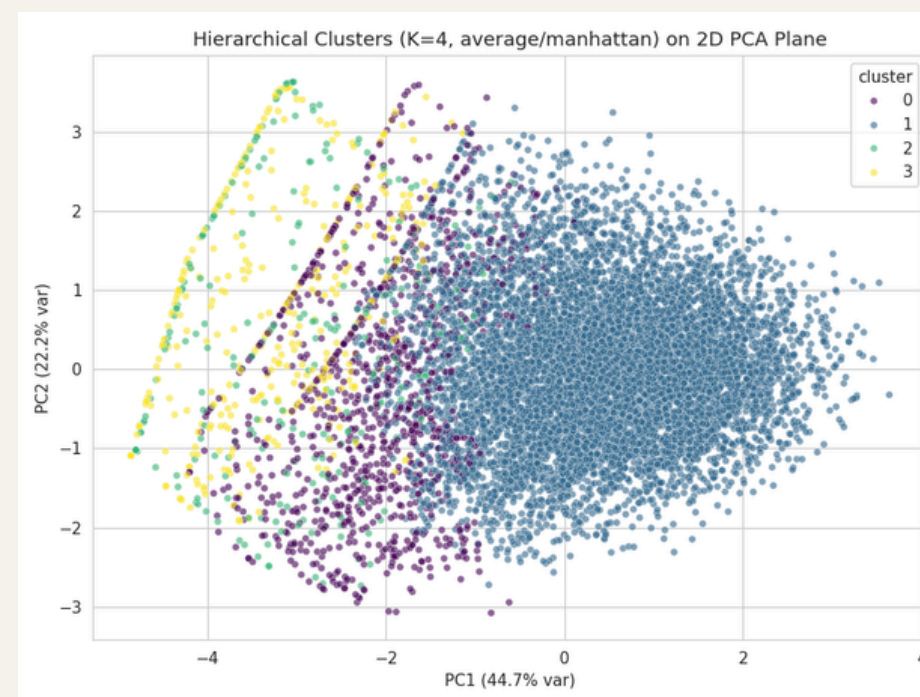
ALGORITHM 02

Hierarchical

- Captures non-spherical clusters; no fixed k initially
- Useful for validating cluster count across methods
- Tuned k = 2–8 · linkage · distance metric

BEST CONFIGURATION

k = 4 · linkage = complete · metric = euclidean

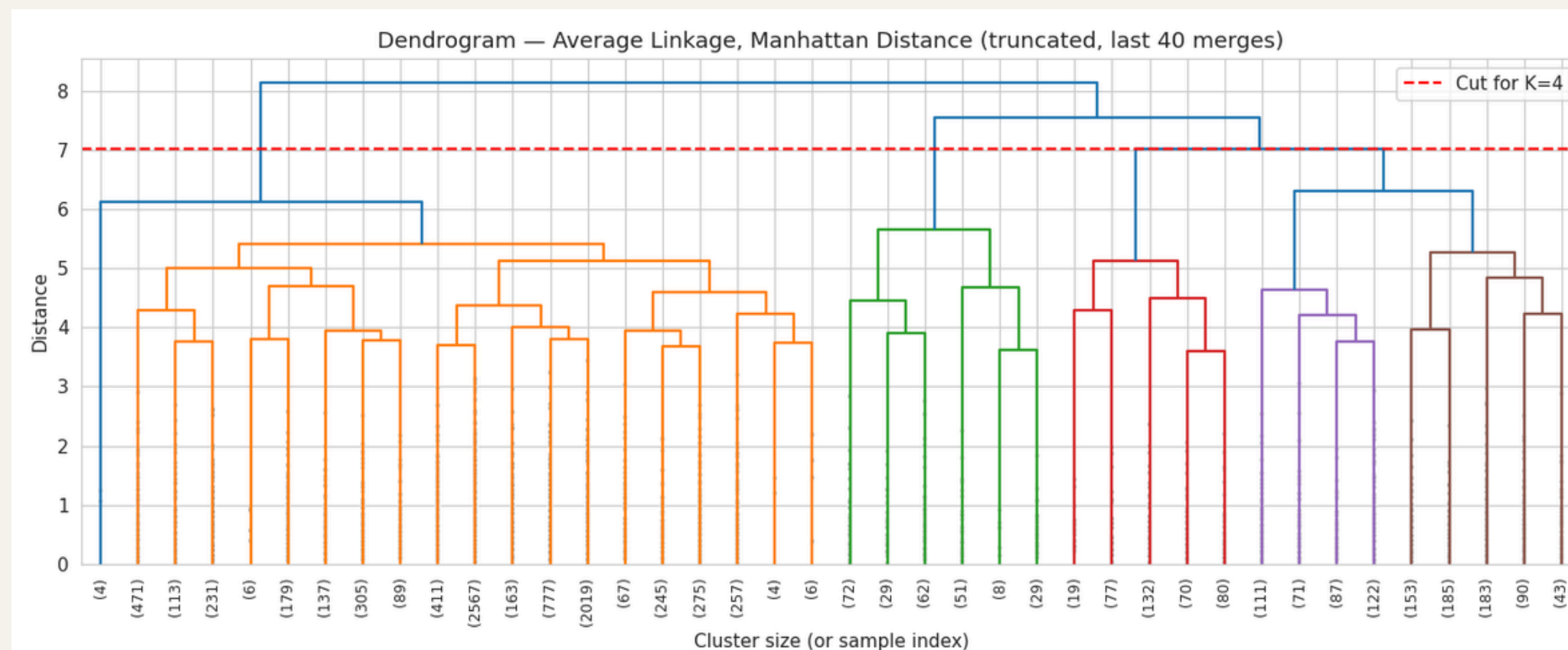
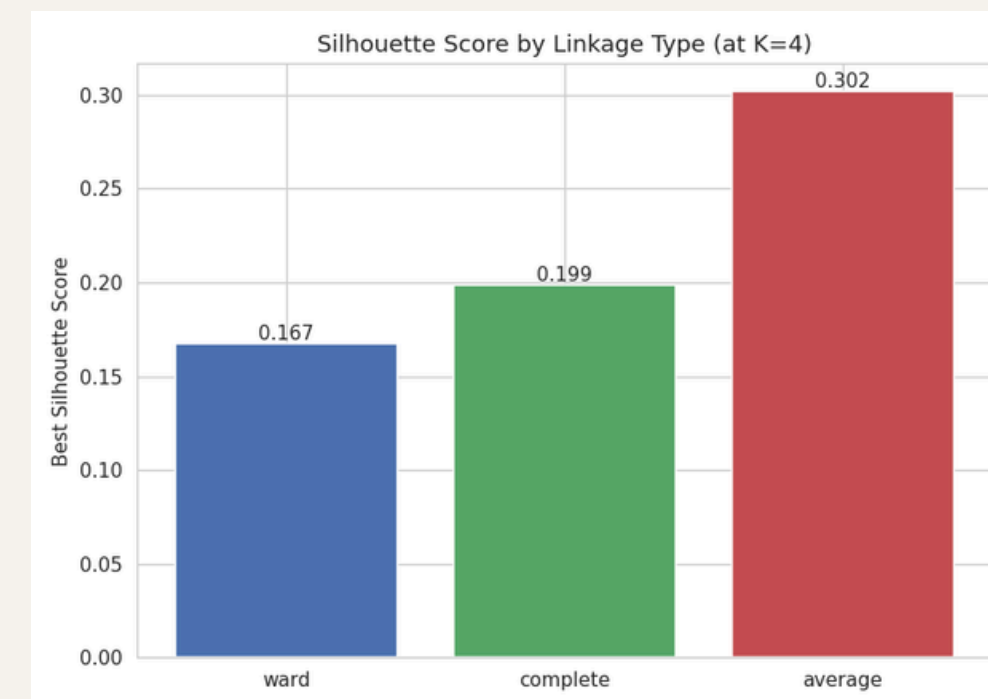
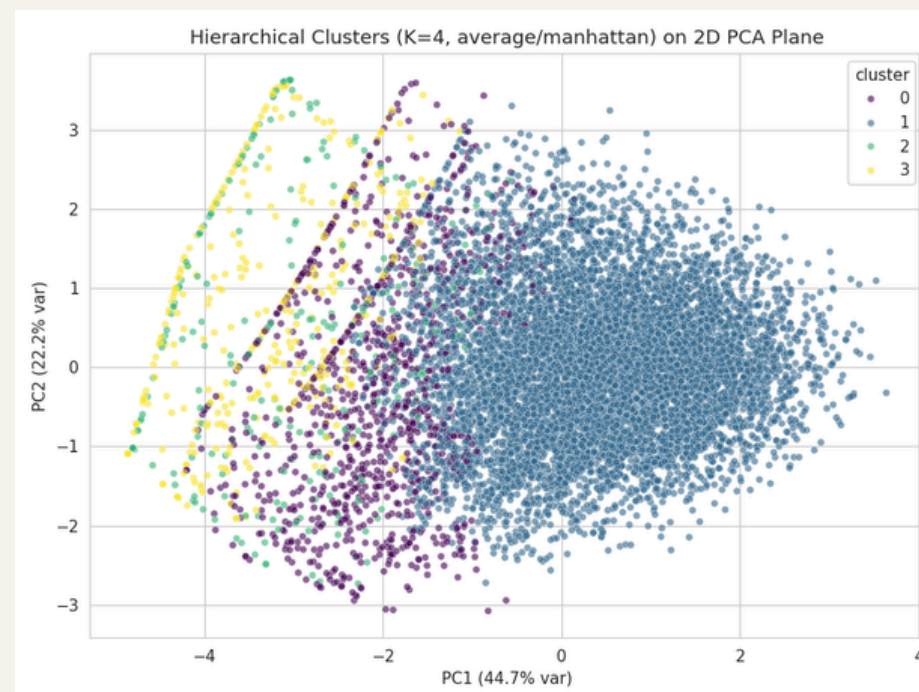


06 • ALGORITHMS

Algorithms Selected & Rationales

Result and Interpretation

- Best Setup: Complete linkage, Euclidean metric (K=4).
- Metrics (Sample = 10000): Silhouette = 0.3016(Highest) | DB = 1.6219 | CH = 1469.2
- Win overall out of the data
- Best when used with K=4



06 • ALGORITHMS

Algorithms Selected & Rationales

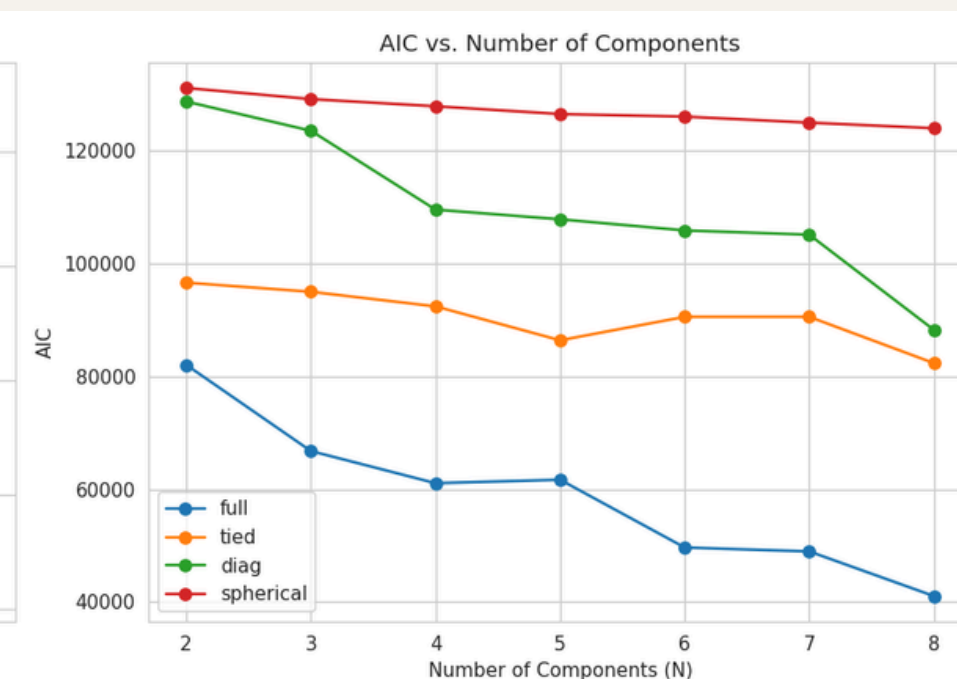
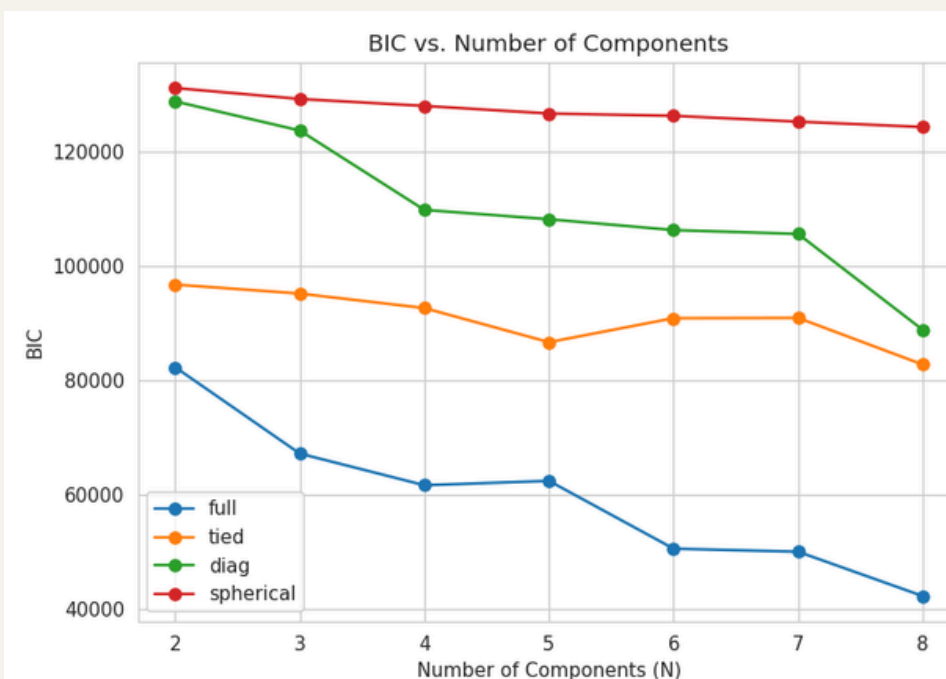
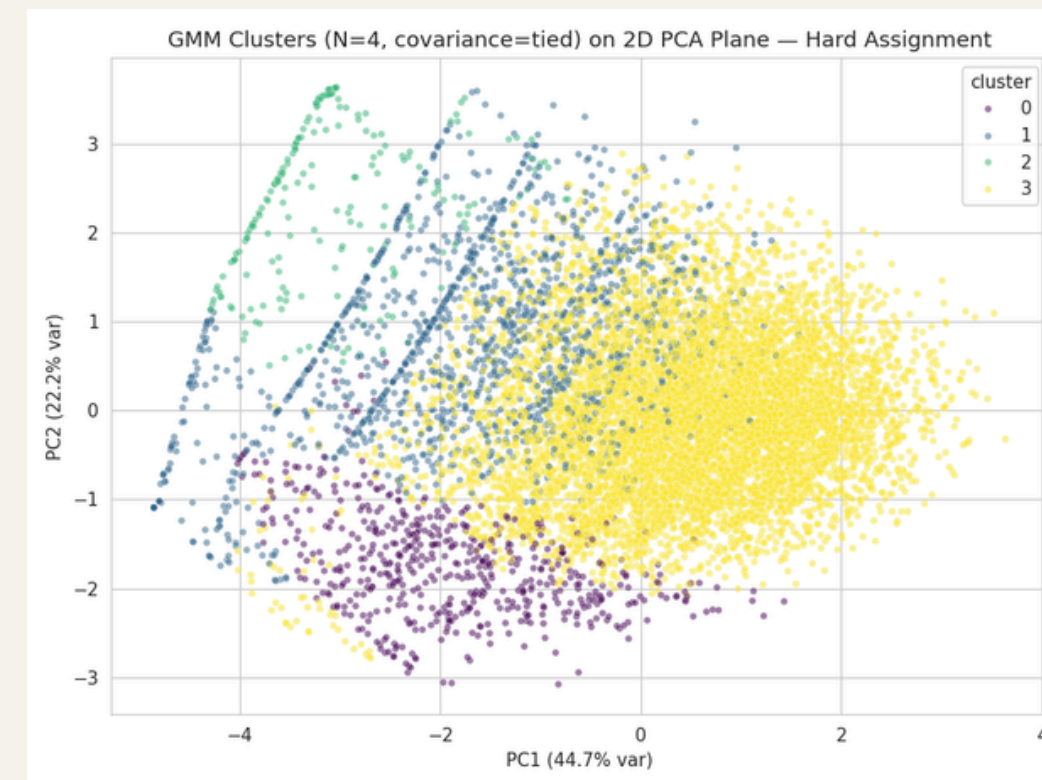
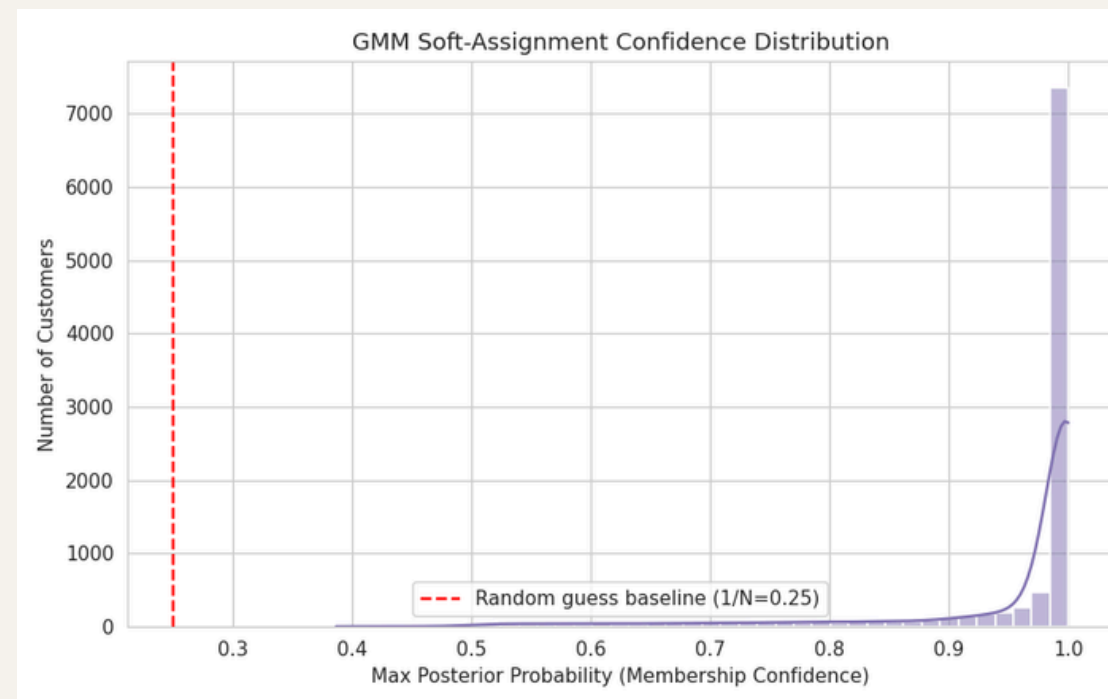
ALGORITHM 03

GMM

- Soft, probabilistic assignment
- Models elliptical (non-isotropic) clusters
- Tuned $n_components = 2-8$ · covariance types

BEST CONFIGURATION

$n_components = 4$ · $covariance = tied$

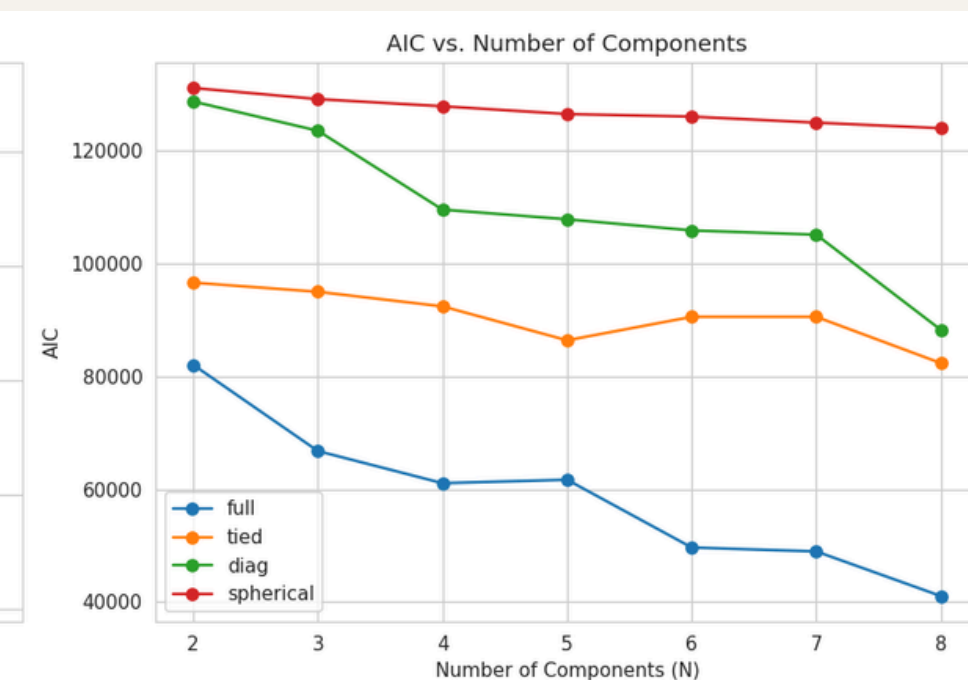
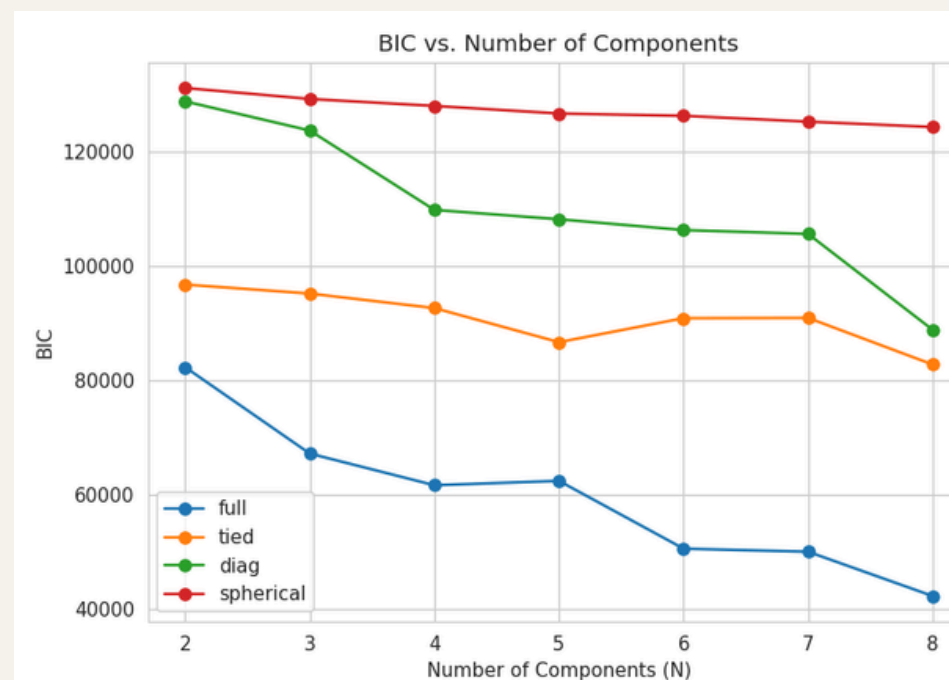
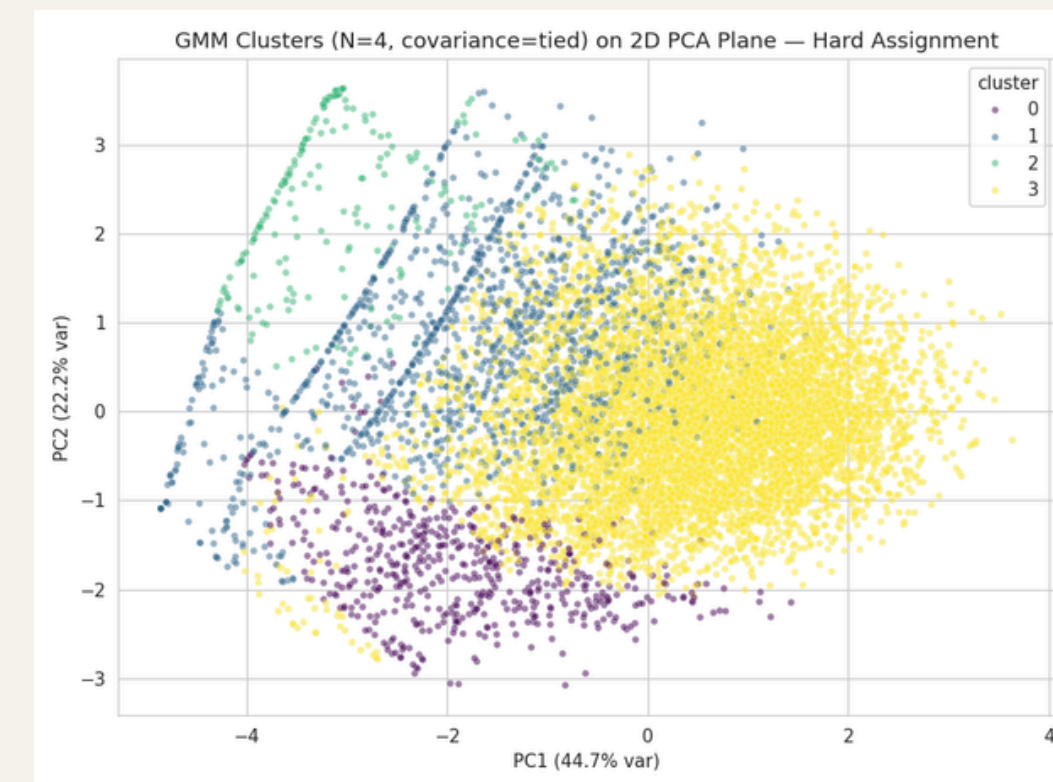
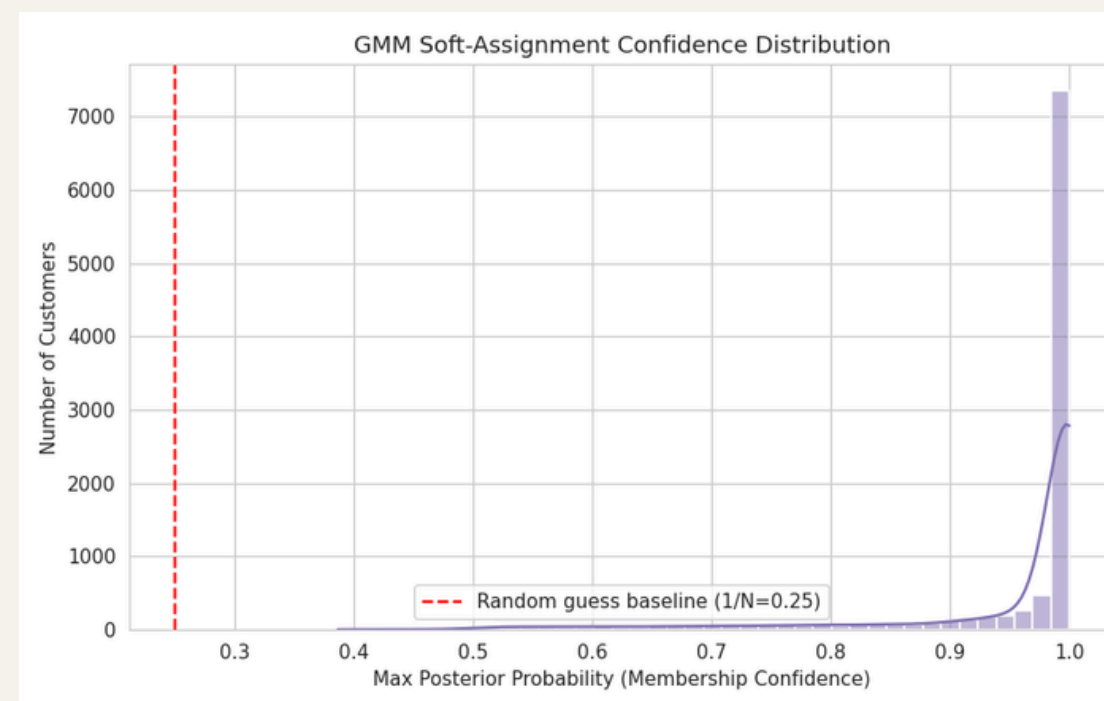


06 • ALGORITHMS

Algorithms Selected & Rationales

Result and Interpretation

- Best Setup: Tied covariance (K=4).
- Metrics: Silhouette = 0.2694(Highest) | DB = 1.4685 | CH = 750.32
- Key Insight: Full covariance won BIC/AIC but dropped Silhouette to 0.159 (overlap). Statistical fit $\neq$ best separation.
- Takeaway: High posterior confidence ($>95\%$) confirms distinct clusters; tied covariance balances flexibility and clean separation.



07 • EVALUATION

Clustering Evaluation Metrics

Silhouette

RANGE -1 to 1 • HIGHER IS BETTER

Measures cluster cohesion vs. separation—the most interpretable geometric metric for clustering quality.



Davies-Bouldin

RANGE ≥ 0 • LOWER IS BETTER

Measures average similarity between nearest clusters—lower values indicate tight, well-separated groups.



Calinski-Harabasz

RANGE ≥ 0 • HIGHER IS BETTER

Ratio of between-cluster to within-cluster dispersion—rewards dense, well-separated partitions.



08 • RESULTS

Side-by-Side Performance

ALGORITHM	BEST PARAMETERS	SILHOUETTE ↑	DAVIES-BOULDIN ↓	CALINSKI-HARABASZ ↑	OUTCOME
K-Means	k=4 • random init • n_init=20	0.2060	1.4648	1438.2640	—
Hierarchical	k=4 • complete linkage • euclidean	0.2996	1.5969	761.0220	WINNER
GMM	n_components=4 • tied covariance	0.2627	1.4685	750.3216	—

KEY TAKEAWAYS

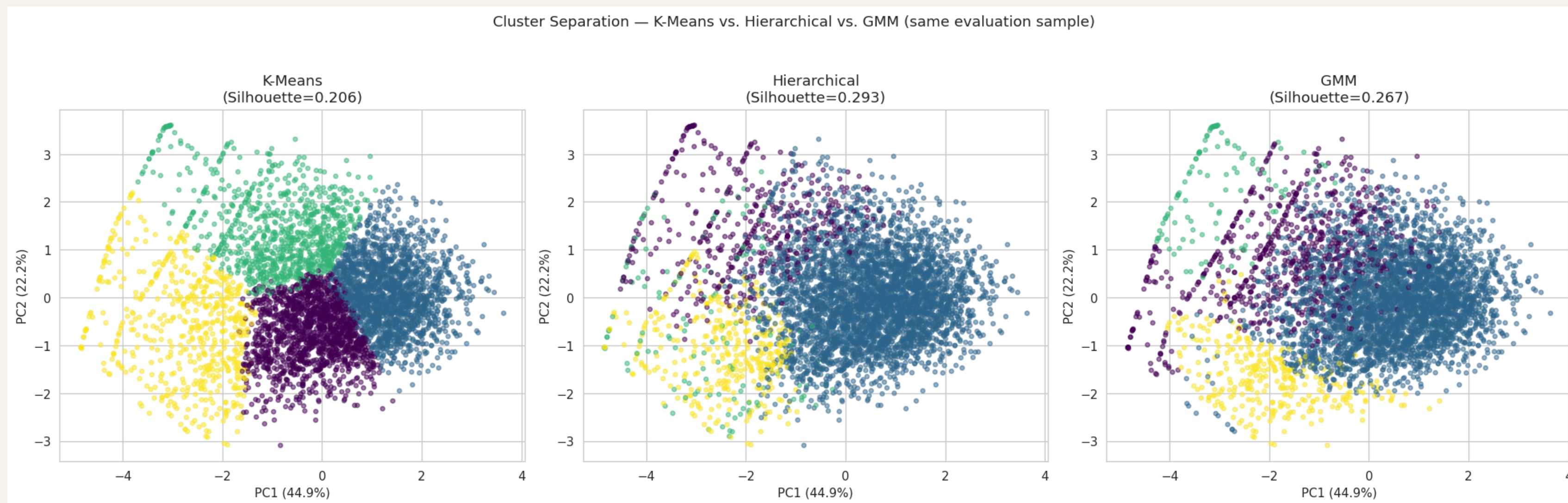
Hierarchical claims the top Silhouette by a clear margin.

CONVERGENCE

All three algorithms independently converge on k = 4.

08 • RESULTS

Visual Cluster Comparison in PCA Space

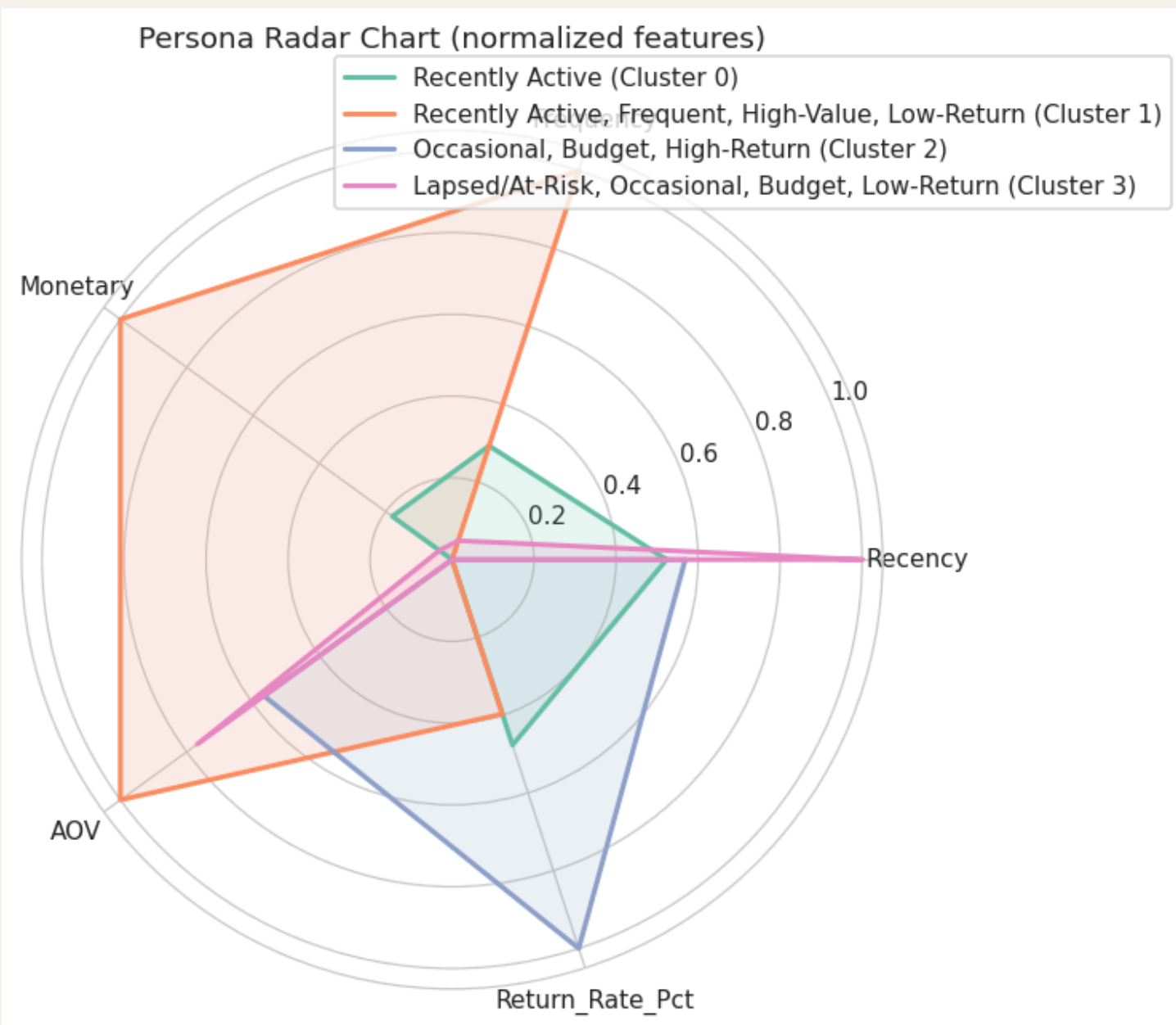


Reading the scatter

Hierarchical clusters appear visibly more compact and better separated than K-Means or GMM. This tighter partition in PCA space corresponds directly to its superior Silhouette score — cohesion within clusters and clear distance between them.

09 • OUTCOME

Best Model & Customer Personas



Four behavioural personas from Hierarchical Clustering.

Recently Active, Frequent, High-Value				40,181	•	81%
RECENCY	FREQ	MONETARY	RETURN %			
207d	5.7	\$15,559	40%			
Lapsed / At-Risk, Occasional				4,593	•	9%
RECENCY	FREQ	MONETARY	RETURN %			
491d	2.3	\$7,643	36%			
Recently Active, Budget, Low-Return				3,629	•	7%
RECENCY	FREQ	MONETARY	RETURN %			
404d	2.7	\$4,318	33%			
Occasional, Budget, High-Return				1,258	•	3%
RECENCY	FREQ	MONETARY	RETURN %			
475d	2.1	\$4,294	97%			

→ The 3% high-return segment may indicate return fraud or persistent product mismatch – a targeted intervention opportunity.

10 • REFLECTION

Why Hierarchical Won & Challenges Faced

WHY HIERARCHICAL WON

Best cohesion, best separation, matching structure.

- 01** Highest Silhouette Score — the strongest cohesion-versus-separation reading of the three.
- 02** Complete linkage produced compact, balanced clusters well-matched to RFM data shape.
- 03** Convergent $k = 4$ cluster count corroborated by K-Means and GMM independently.

CHALLENGES FACED

Trade-offs, caveats, honest limitations.

Scalability

$O(n^2)$ memory forced Hierarchical onto a 4,000-customer sample, reducing comparison fairness on absolute metrics.

Deployment

Hierarchical does not predict for new customers — required a surrogate k-NN model to score incoming users.

Preprocessing

RFM preprocessing (log, cap, standardize) was critical due to skew and outliers.

Modest scores

Silhouettes of 0.20–0.29 reflect natural overlap in real customer behaviour, not weakness of method.

10 • REFLECTION

Future Plan after this project

Model Deployment & Monitoring

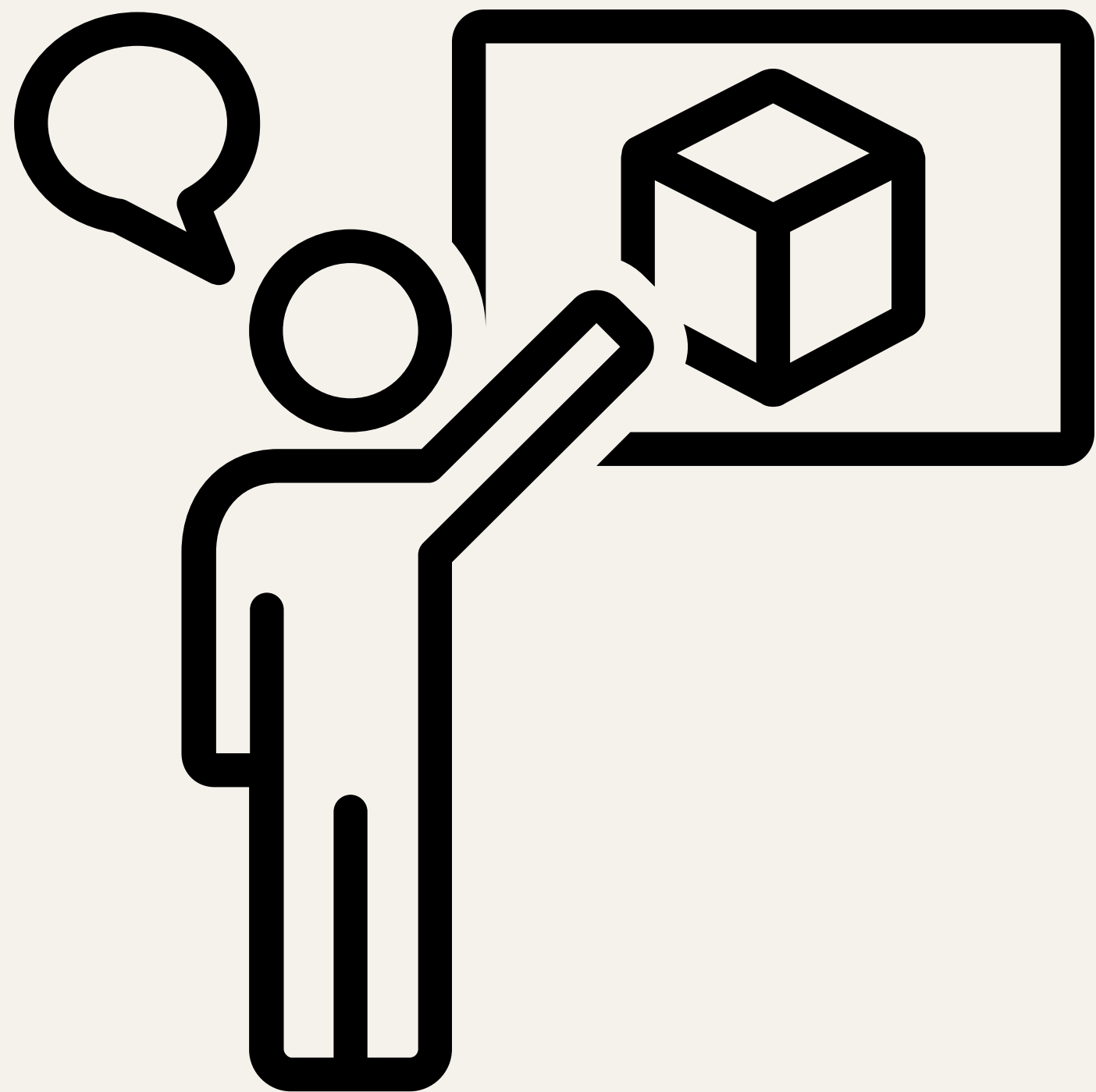
- Production Integration: Deploy the final model via API/pipeline to dynamically segment incoming users.
- Drift Monitoring: Continuously track cluster stability (DB, CH metrics) to detect shifting behaviors and data drift.

Business Activation & Marketing Strategy

- Personalized Campaigns: Tailor messaging, dynamic recommendations, and promotions based on cluster personas.
- Product Alignment: Adapt features and onboarding flows to match segment-specific behaviors.

Advanced Modeling & Future Iterations

- Classifier Pipeline: Train an XGBoost/Random Forest model on cluster labels to score new users immediately.
- Dynamic Modeling: Upgrade static RFM to time-series models (RNNs/Markov chains) for tracking customer lifecycle shifts.
- A/B Testing: Validate ROI and conversion lift by testing segment-targeted strategies against baselines



Demo of ML



Questions??